

STARTMED

Telemedicine Application

Team 2

1. Task Division

Team Role	Responsibilities
Project Manager	Oversees timeline, coordinates phases, maintains documentation, and produces ER Diagrams and system design artefacts.
Frontend Developer	Builds the React.js Patient Portal: registration UI, profile management, appointment booking screens, and responsive layouts.
Backend Developer	Implements Python REST APIs, JWT authentication, role-based access control (Admin / Clinic Staff / Patient), and server-side business logic.
Database Administrator	Designs the MySQL schema, produces the ER Diagram, manages migrations, and ensures data integrity for medical records.
Full-Stack / Integration Lead	Integrates appointment management module, billing/payment processing, and WebRTC video consultation pipeline.

X2. System Design

2.1 System Architecture Overview

STARTMED follows a three-tier client-server architecture:

- **Presentation Layer:** React.js Single-Page Application (SPA) served to the browser, communicating with the backend via RESTful APIs.
 - **Application Layer:** Python (Flask/FastAPI) backend handling business logic, authentication, and real-time WebRTC signalling.
 - **Data Layer:** MySQL relational database storing patient records, appointments, billing data, and user credentials.
-

2.2 Core Modules

Feature / Module	Description
Patient Registration & Login	Patients self-register; JWT tokens issued on login; role assigned (Patient).
Doctor / Staff Portal	Clinic staff and admins log in via separate portal; access dashboards based on role.
Appointment Scheduling	Patients browse available slots; book, reschedule, or cancel appointments.
Video Consultation (WebRTC)	Real-time encrypted video/audio sessions between doctor and patient.
Electronic Health Records	Doctors create, update, and view patient records; history and prescriptions stored.
Digital Billing	Auto-generated invoices post-consultation; online payment integration.
Admin Dashboard	Manage users, view analytics, configure system settings.

2.3 User Roles

- **Admin**
 - Full system access: manage users, view analytics, configure settings.
- **Clinic Staff / Doctor**
 - Access patient records, manage appointments, conduct video consultations, write prescriptions.
- **Patient**
 - Register, book appointments, join video consultations, view own medical history and invoices.

2.4 Technology Stack

The stack selected in Phase 0 remains unchanged:

- Frontend: React.js — component-based SPA, responsive design.
- Backend: Python (Flask or FastAPI) — RESTful API server.
- Database: MySQL — relational schema for structured medical data.
- Authentication: JSON Web Tokens (JWT) with role-based access control.
- Communication: WebRTC — browser-native, peer-to-peer encrypted video/audio.

3. ER Diagram

3.1 Entities & Attributes

The following entities form the core of the STARTMED data model:

- **User**
 - user_id (PK), full_name, email, password_hash, role (Admin | Staff | Patient), created_at
- **Patient**
 - patient_id (PK), user_id (FK), date_of_birth, gender, phone, address, blood_type
- **Doctor**
 - doctor_id (PK), user_id (FK), specialty, license_number, available_slots
- **Appointment**
 - appointment_id (PK), patient_id (FK), doctor_id (FK), scheduled_at, status (Pending | Confirmed | Cancelled | Completed), notes
- **Medical Record**
 - record_id (PK), patient_id (FK), doctor_id (FK), appointment_id (FK), diagnosis, prescription, created_at
- **Invoice**
 - invoice_id (PK), appointment_id (FK), patient_id (FK), amount, status (Unpaid | Paid), issued_at, paid_at
- **Consultation Session (WebRTC)**
 - session_id (PK), appointment_id (FK), room_token, started_at, ended_at, duration_minutes

4. Feature Choosing & Assigning

4.1 Selected Features for Phase 1

- Secure user registration and JWT-based login for all roles.
 - Role-based dashboard routing (Admin / Doctor / Patient).
 - Patient profile creation and editing.
 - Doctor profile and available-slot management.
-

- Appointment booking, confirmation, and cancellation flow.
- WebRTC video consultation room (basic session initiation).

4.2 Feature Assignment Matrix

Feature
Authentication & JWT
Patient Portal UI
Appointment Management
Database Schema & ER Diagram
WebRTC Video Consultation
Electronic Health Records
Billing Module
Admin Dashboard
System Design Documentation
